

Table A-1 Zeros and Weights for Gauss-Legendre Quadrature

n	$\pm \xi_k$	w_k	n	$\pm \xi_k$	w_k
2	0.5773502692	1.0000000000	12	0.1252334085	0.2491470458
3	0.0000000000	0.8888888889		0.3678314990	0.2334925365
	0.7745966692	0.5555555556		0.5873179543	0.2031674267
4	0.3399810436	0.6521451549		0.7699026742	0.1600783285
	0.8611363116	0.3478548451		0.9041172564	0.1069393260
5	0.0000000000	0.5688888889	16	0.9815606342	0.0471753364
	0.5384693101	0.4786286705		0.0950125098	0.1894506105
	0.9061798459	0.2369268850		0.2816035508	0.1826034150
				0.4580167777	0.1691565194
6	0.2386191861	0.4679139346		0.6178762444	0.1495959888
	0.6612093865	0.3607615730		0.7554044084	0.1246289713
	0.9324695142	0.1713244924		0.8656312024	0.0951585117
7	0.0000000000	0.4179591837	20	0.9445750231	0.0622535239
	0.4058451514	0.3818300505		0.9894009350	0.0271524594
	0.7415311856	0.2797053915		0.0765265211	0.1527533871
	0.9491079123	0.1294849662		0.2277858511	0.1491729865
8	0.1834346425	0.3626837834		0.3737060887	0.1420961093
	0.5255324099	0.3137066459		0.5108670020	0.1316886384
	0.7966664774	0.2223810345		0.6360536807	0.1181945320
	0.9602898565	0.1012285363		0.7463319065	0.1019301198
9	0.0000000000	0.3302393550		0.8391169718	0.0832767416
	0.3242534234	0.3123470770		0.9122344283	0.0626720483
	0.6133714327	0.2606106964		0.9639719273	0.0406014298
	0.8360311073	0.1806481607	24	0.9931285992	0.0176140071
	0.9681602395	0.0812743884		0.0640568929	0.1279381953
10	0.1488743390	0.2955242247		0.1911188675	0.1258374563
	0.4333953941	0.2692667193		0.3150426797	0.1216704729
	0.6794095683	0.2190863625		0.4337935076	0.1155056681
	0.8650633667	0.1494513492		0.5454214714	0.1074442701
	0.9739065285	0.0666713443		0.6480936519	0.0976186521
				0.7401241916	0.0861901615
				0.8200019860	0.0733464814
				0.8864155270	0.0592985849
				0.9382745520	0.0442774388
				0.9747285560	0.0285313886
				0.9951872200	0.0123412298

This table of zeros and weights for Gauss-Legendre quadrature has been excerpted from Ref. 15 and rounded to 10 decimal places. More extensive tables can also be found in Ref. 14.